UNIT 4 · TOPIC 7(A)

Emissions and Control of Pollutants

Syllabus topic: Recent trends in tractor design; emissions and control of pollutants

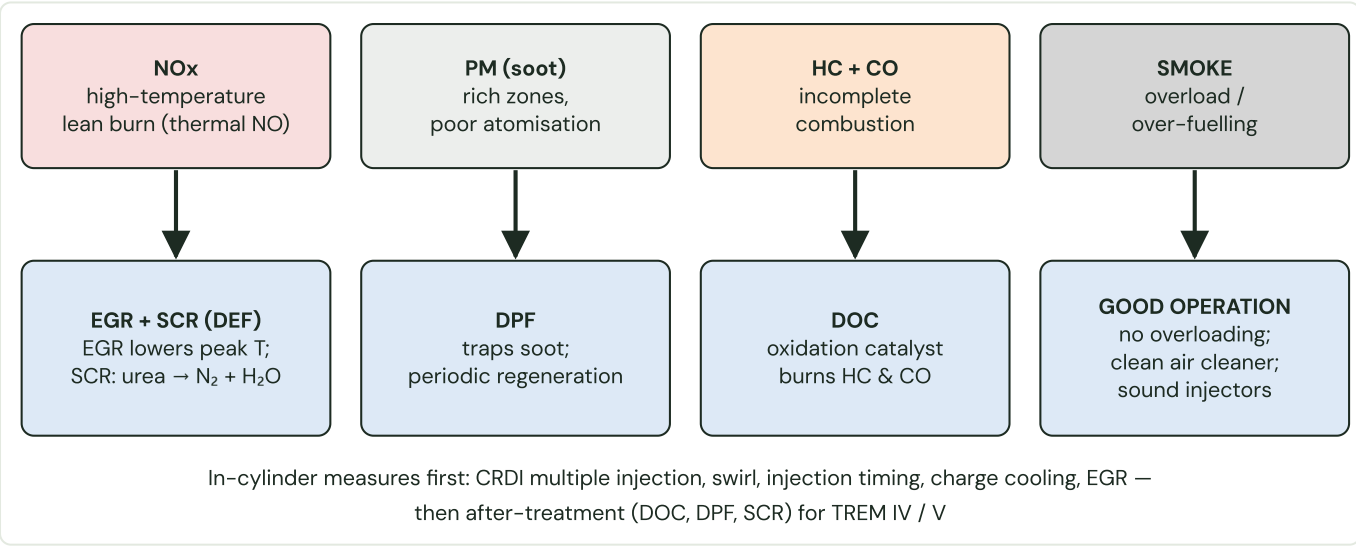
Diesel pollutants

Pollutant	Origin
NOx	Thermal — high combustion temperatures in lean mixtures (excess O_2)
Particulate matter (PM) / soot	Locally rich zones, poor atomisation
HC and CO	Incomplete combustion — small in diesels
Visible smoke	Overload

The NOx–PM trade-off is the central difficulty: measures that cool combustion cut NOx but raise PM.

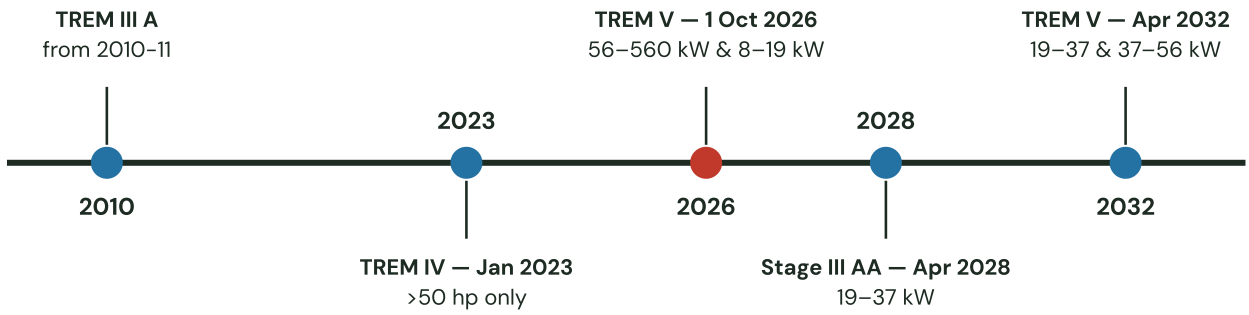
Control chain — in-cylinder first, then after-treatment

1. **In-cylinder:** retarded/shaped injection, CRDI high-pressure multiple injections, swirl/chamber design, charge cooling, **EGR** (recirculated exhaust dilutes O_2 and lowers peak temperature → less NOx).
2. **DOC** (diesel oxidation catalyst): oxidises HC and CO.
3. **DPF** (diesel particulate filter): traps soot; needs periodic regeneration.
4. **SCR** with urea-DEF (AdBlue): reduces NOx to N_2 + H_2O.



Diesel pollutants and their control (redrawn — the source figure paired smoke with SCR, which treats NOx)

Indian tractor emission norms (TREM)



PM limit tightens 0.025 → 0.015 g/kWh; TREM V aligns with EU Stage V

(dates as given in the source notes, July 2026 — confirm the final gazette notification)

Tractor (TREM) emission norms roadmap (redrawn; the source figure's title was clipped)

Stage	Applies to	From
TREM III A	Tractors	2010-11
TREM IV	>50 hp tractors only	January 2023
TREM V	56–560 kW tractors and combines, AND 8–19 kW engines	1 October 2026
Stage III-AA → TREM V	19–37 kW	April 2028 → April 2032
TREM V	37–56 kW	April 2032

- PM limit tightens **0.025 → 0.015 g/kWh**; TREM V aligns with **EU Stage V**, pushing DPF/SCR after-treatment onto larger tractors.
- The Tractor Manufacturers Association (TMA) sought deferral for the 25–50 hp class (Aug 2025) — track the final notification.

Status note: dates are as given in the source notes (verified July 2026). Reports during 2026 differ on whether the final gazette notification fixed these dates — check the MoRTH notification (egazette.gov.in) before the exam.

What after-treatment means for farms

- After-treatment on larger tractors: EGR, DOC, DPF (PM 0.015 g/kWh under TREM-V) and SCR with DEF/AdBlue dosing.
- Maintenance: DEF storage/refill, DPF regeneration cycles, low-ash (CK-4 class) oils, stricter fuel cleanliness (10 ppm S BS-VI).

Practice MCQs

Q32. TREM-IV norms (Jan 2023) applied to tractors of:

(a) All sizes
(b) Above 50 hp only
(c) Below 20 hp
(d) Combines only

Q33. TREM-V norms for 56–560 kW tractors/combines take effect from:

(a) Jan 2023
(b) 1 Oct 2026
(c) 2030
(d) 2035

Q34. Under TREM-V the PM limit for major categories tightens to:

(a) 0.15 (b) 0.075

(c) 0.025 (d) 0.015 g/kWh

Q35. TREM-V aligns Indian tractor emissions with:

(a) US Tier 2 (b) EU Stage V

(c) BS-III (d) China-IV

Q193. The two chief diesel pollutants in tension with each other are:

(a) CO and HC (b) NOx and PM

(c) O_2 and N_2 (d) SO_2 and CO_2

Q194. Thermal NOx forms mainly due to:

(a) Low temperature (b) High combustion temperature with excess O_2

(c) Rich mixtures (d) Cold starts

Q195. EGR reduces NOx by:

(a) Adding oxygen (b) Lowering peak combustion temperature with recirculated exhaust

(c) Filtering (d) Urea

Q196. Diesel particulate matter is captured by the:

(a) DOC (b) DPF

(c) EGR (d) Muffler

Q197. SCR systems reduce NOx using:

(a) Petrol (b) Urea solution (DEF/AdBlue)

(c) Water (d) Oil

Q198. The DOC primarily oxidises:

(a) NOx (b) HC and CO

(c) N_2 (d) Soot only

Q199. TREM-IV norms (Jan 2023) applied to tractors:

(a) Of all sizes (b) Above 50 hp

(c) Below 20 hp (d) Only combines

Q200. TREM-V starts for 56–560 kW tractors and combines from:

(a) Apr 2020 (b) Jan 2023

(c) 1 Oct 2026 (d) 2032

Q201. Under TREM-V the PM limit for major categories becomes:

(a) 0.15 (b) 0.025

(c) 0.015 (d) 0.005 g/kWh

Answers

Q32 — (b) Above 50 hp only

Q33 — (b) 1 Oct 2026 — *As given in the source notes (verified July 2026) — watch for further TREM-V deferral news for the 25–50 hp class.*

Q34 — (d) 0.015 g/kWh

Q35 — (b) EU Stage V

Q193 — (b) NOx and PM

Q194 — (b) High combustion temperature with excess O_2

Q195 — (b) Lowering peak combustion temperature with recirculated exhaust

Q196 — (b) DPF

Q197 — (b) Urea solution (DEF/AdBlue)

Q198 — (b) HC and CO
Q199 — (b) Above 50 hp
Q200 — (c) 1 Oct 2026 — *TREM dates as given in the source notes (MoRTH notifications, verified July 2026).*
Q201 — (c) 0.015

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units; standard thermodynamics texts; PIB/MoPNG releases on ethanol blending; MoRTH/CMVR TREM notifications and DieselNet (as verified in the source notes, July 2026); NPTEL lectures. Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units (2nd ed., 1989), Ch. 14 "Tractor Tests and Performance", pp. 403–446; MoRTH TREM notifications (as verified in the source notes, July 2026); CFMTTI Budni documentation; Jagdishwar Sahay (maintenance schedules); NPTEL lectures.